MILITARY SPECIFICATION

HOSE AND FITTINGS, OIL AND TURBINE FUEL, SMOOTH BORE, CORRUGATED COVER, 7-INCH

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 7-inch, smooth bore hose with corrugated cover, including fittings, used for receiving oil and turbine fuel with the probe fueling system receivers.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
TT-S-735 - Standard Test Fluids; Hydrocarbon.

MILITARY
MIL-H-775 - Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers, Packaging of.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-24356 - Couplings, Segmented, Reattachable, 4-, 6-, and 7-Inch, Adapters and Accessories for Refueling-At-Sea Hose.

STANDARDS

FEDERAL
FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.
FED-STD-601 - Rubber; Sampling and Testing.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B221 - Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes.
B247 - Aluminum-Alloy Die and Hand Forgings.
D380 - Rubber Hose, Testing.
D381 - Existent Gum in Fuels by Jet Evaporation, Test for.
D412 - Rubber Properties in Tension, Tests for.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Construction. Hose dimensions shall conform dimensionally to figure 1. The hose shall have a rubber tube, reinforcing of nylon or polyester (applied as fabric, braid, or spiral wrap), a helix of round wire, additional nylon or polyester reinforcing, and a rubber cover.

3.2 Wire. The helix wire shall be high strength metal, treated if necessary to prevent corrosion, having an outside diameter (o.d.) of 0.207 inch. The helix shall be applied in the direction of a right hand thread having a stitch of 1 inch and shall be continuous to both ends of the hose.

3.3 Length. Hose lengths, for bulk hose, shall be 8 feet 8 inches ± 1/2-inch long. Hose may be fabricated in longer lengths to include samples for testing. Not less than 5 hoses, for each contract or order for 5 or more hoses, shall be fabricated longer than the 8 feet 8 inch ± 1/2-inch specified length to provide samples for the tests specified in 4.6 through 4.6.9.9.3.

3.4 Tube and cover thickness. Thickness of the tube and cover shall be a minimum of 0.063 inches after buffing.

3.5 Diameter. The hose shall have a 7-inch nominal inside diameter (i.d.).

3.6 Weight. The maximum weight of the hose shall be 11.0 pounds per foot.

3.7 End fittings. End fittings shall be fabricated of aluminum in accordance with ASTM B221 or ASTM B247, and anodized in accordance with type I, MIL-A-8625. End fittings shall be reuseable. The socket shall be of the segmented type and the segments shall clamp over the spiral wire reinforcement helix in the hose carcass without special construction on the hose ends. Hose fitting and split clamp are shown on figures 2, 3, and 4.

3.8 Flexibility. One length of coupled hose as specified in contract or order (see 6.2.1) shall be tested as specified in 4.6.10. Bending and torquing the hose shall not reduce the minor diameter of the resulting elipse to less than 5 inches. The wire helix shall not break or break through the tube or cover of the hose.

3.9 Burst pressure. When tested as specified in 4.6.7, the hose shall not leak or burst, nor shall the end fittings come off the hose at up to 500 pounds per square inch (lb/in^2) pressure.

3.10 Proof pressure. When tested as specified in 4.6.5 with a proof pressure of 250 lb/in^2, there shall be no leakage or damage to the hose as a result of this test.

3.10.1 Dimensional stability. The hose shall not increase in length more than 8 percent nor shall the hose twist more than 1 degree per foot of hose when subjected to the proof test of 4.6.5.

3.11 Rubber components. The properties of the rubber in the tube, cover, and friction of the hose shall conform to table I. Recommended formulations for these components are given in 6.3. The rubber used in the tube and friction shall be a copolymer of butadiene and acrylic nitrite. The rubber used in the cover shall be a blend of a copolymer of butadiene and acrylic nitrite and polyvinylchloride.

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TABLE I. Requirements for rubber components in the hose.

	Tube	Cover	Friction	Test paragraph
Initial properties:				
Tensile strength, lb/in^2, minimum	2600	1600		4.6.9.1
Ultimate elongation	300	350		4.6.9.1
Adhesion, pounds per inch, minimum	15	15	15	4.6.9.5
Properties after immersion in test fluid:				
Tensile strength, lb/in^2, minimum	1400	800		4.6.9.2
Ultimate elongation, percent minimum	200	200		4.6.9.2
Volume increase, percent maximum (no shrinkage permitted)	30	30	30	4.6.9.4
Adhesion, pounds per inch, minimum	10	10	10	4.6.9.6
Properties after oven aging:				
Tensile strength, lb/in^2, minimum	2000	1450		4.6.9.3
Ultimate elongation, percent minimum	250	250		4.6.9.3
Nonvolatile extract from tube, percent maximum	3.0			4.6.9.7
Phosphate plasticizer in tube, qualitative determination	Absent			4.6.9.8
Resistance of cover to ozone		No cracking		4.6.9.9 through 4.6.9.9.3

3.12 Hose and end fitting marking.

3.12.1 Hose. Hose shall have 2 rubber labels vulcanized into the cover, with one label located 2 feet from each end. Labels shall contain the following:

(a) Manufacturer's name or trademark.
(b) Month and year of manufacture.
(c) Specification number.
(d) "7-inch receiver hose".

3.12.2 End fittings. End fittings shall be stamped with the following:

(a) Manufacturer's name or trademark.
(b) Specification number.
(c) Part number, manufacturers.
(d) Nipples and segments shall contain the manufacturer's piece number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examination in 4.5 and tests specified in 4.6 through 4.6.10.

4.3 Sampling for quality conformance inspection.

4.3.1 Sampling for examination, measurements and tests performed on entire hose. For the purposes of sampling, the lot shall consist of all hose manufactured under the same conditions and which are offered for delivery at one time. A random sample of hoses shall be selected from each lot in accordance with table II and subjected to the examination and tests specified in 4.5 and 4.6.1 through 4.6.6.

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TABLE II. Sampling for examination, measurements and tests performed on entire hose. AQL (approx.) = 2.5 percent defective for major defects and 10.0 percent defective for major and minor defects combined.

Number of hoses in lot	Number of hoses in sample	Major defects		Total defects	
		Acceptance number (defectives)	Rejection number (defectives)	Acceptance number (defectives)	Rejection number (defectives)
15 and under	10	0	1	2	3
16 to 40	15	1	2	3	4
41 to 110	25	1	2	5	6
111 to 300	35	2	3	7	8
301 and over	50	3	4	9	10

4.3.2 Sampling for the measurements and tests performed on sections cut from hose. The extra length shall be cut from the 5 extra long hoses (see 3.3) and specimens prepared as required for the tests specified in 4.6.7 through 4.6.9.9.3.

4.3.2.1 After cutting the extra lengths from the 5 extra long hoses, the residual length of the remaining hose shall be 8 feet 8 inches ± 1/2-inch.

4.3.3 Sample for swelling test of rubber friction. A vulcanized sample of the rubber friction used to impregnate the fabric, braid or spiral wrap reinforcement shall be provided for the friction swelling test (see 4.6.9.4). The sample shall be 6 inches by 6 inches by 0.080 inch ± 0.010 inch, and shall be accompanied by written certification from the contractor that the sample has the same composition as the friction used in the hose and an equivalent cure.

4.3.4 Sample for flexibility test. One hose shall be selected from each contract or order and subjected to the flexibility test of 4.6.10.

4.4 Rejection.

4.4.1 Rejection on basis of visual examination and measurements of length. The lot shall be rejected when the number of defective hoses exceeds the acceptance number specified in table II. In any case, every hose containing one or more major defects shall be rejected and counted against the acceptance number for defective hoses. Every hose containing more than three minor defects shall be rejected and counted against the acceptance number for total defects. Every hose containing three or less minor and no major defects shall be counted a minor defective against the acceptance number for the total defectives, but shall not be rejected.

4.4.1.1 Classification of visual defects and defects in length. These defects shall be classified in accordance with the following:

(a) Visual defects shall be classified in accordance with FED-STD-162.
(b) Length defects, hoses shorter than specified (see 3.3), shall be rejected as a major defect. Hoses which are too long shall be cut to the specified length.

4.4.2 Rejection on basis of wall configuration. The lot shall be rejected if the wall configuration is other than that shown on figure 1.

4.4.3 Rejection on the basis of other tests performed on entire hose. The lot shall be rejected if any sample hose (see 4.3.1) fails to meet the requirements for unit weight (see 3.6), proof pressure (see 3.10) or dimensional stability (see 3.10.1).

4.4.4 Rejection on basis of tests performed on hose sections and rubber samples. The lot shall be rejected if the hose sections (see 4.3.2) fail to meet the requirements for burst pressure (see 3.9), tube or cover thickness (see 3.4) or properties of rubber compounds (see 3.11).

4.5 Visual examination. The interior and exterior surfaces of sample hoses (see 4.3.1) shall be examined for the visual defects described in FED-STD-162.

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4.6 Tests. Tests shall be performed at 80°F ± 9°F, except where otherwise specified herein.

4.6.1 Length. Lengths of sample hoses (see 4.3.1) shall be measured as specified in ASTM D380.

4.6.2 Inside diameter. The i.d. of sample hoses (see 4.3.1) shall be measured as specified in ASTM D380 using plug gages.

4.6.3 Wall configuration and thickness. Wall configuration and thickness of sample hoses (see 4.3.1) shall be measured by vernier caliper and template or suitable dial gage. Measurements shall be made at both ends of the hose.

4.6.4 Unit weight. Unit weights of sample hoses (see 4.3.1) shall be determined by weighing hoses without couplings. Readings shall be accurate to the closest 1/2-pound.

4.6.5 Proof pressure. Sample hoses (see 4.3.1) shall be equipped with couplings and subjected to the proof pressure specified in 3.10. Before starting the test, a 10-pound hydrostatic pressure shall be established in the hose, and the length, o.d., and zero index for the twist shall be measured (see ASTM D380) in preparation for the dimensional stability test specified in 4.6.6. The method shall be as described in ASTM D380 except that the hydrostatic pressure rate shall not exceed 1000 lb/in^2 per minute. The proof pressure shall be held for 5 minutes and the hose and fittings examined for leakage or other indication of weakness. At the end of the 5 minute period and while the pressure is maintained, the length, o.d., and twist of the hose shall be measured for the determination of dimensional stability.

4.6.6 Dimensional stability. The effect of the proof test pressure (see 3.10) on the length, o.d., and twist of sample hoses (see 4.3.1) shall be determined as specified in 3.10.1 and ASTM D380. The gage pressure shall then be reduced to zero lb/in^2, and the hose left in this condition for 5 minutes. At the end of this period, the diameter and length of the hose shall be measured.

4.6.7 Burst pressure. The hose section obtained in accordance with 4.3.2 shall be equipped with couplings in accordance with MIL-C-24356. The burst pressure of this section in the straight condition shall then be determined in accordance with ASTM D380.

4.6.8 Tube and cover thicknesses. Thicknesses of the tube and cover shall be measured on specimens removed from each end of the hose section obtained in accordance with 4.3.2. Thicknesses shall be determined by the procedure given in ASTM D380. Respective values found for the tube thickness shall not be averaged, but shall be regarded as independent measurements, each of which should meet the requirement. Respective values for the cover thicknesses shall be treated in the same manner.

4.6.9 Properties of rubber components. Properties of rubber conponents, except the swelling in testing fuel of the friction rubber on the reinforcing fabric, shall be determined on longitudinal specimens prepared from the hose section obtained in accordance with 4.3.2. After separating and buffing smooth, the tube and cover samples shall be permitted to rest not less than 30 minutes before preparing test specimens. Swelling in testing fuel of the friction rubber shall be determined using specimens cut from the sample obtained in accordance with 4.3.3.

4.6.9.1 Initial tensile properties. Initial tensile strength and ultimate elongation shall be determined in accordance with ASTM D412.

4.6.9.2 Tensile properties after immersion. Tensile strength and ultimate elongation after immersion in type III fluid of TT-S-735 shall be determined in accordance with method 6111 of FED-STD-601, using specimens cut with die III. Tensile strength shall be based on the swollen cross-section area of the specimen.

4.6.9.3 Tensile properties after oven aging. Tensile strength and ultimate elongation after oven aging shall be determined in accordance with method 7221 of FED-STD-601, using specimens cut with die III, except that the aging period shall be 70 hours ± 1/4 hour at 212°F ± 2°F.

4.6.9.4 Volume change after immersion. Change in volume in percent of the original volume after immersion in type III fluid of TT-S-735, shall be determined on samples of the tube, cover, and friction in accordance with method 6211 of FED-STD-601.

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4.6.9.5 Initial adhesion. Adhesion of the tube and cover of the reinforcement, and of the plies of reinforcement to each other, shall be determined in accordance with method 6411 of FED-STD-601, except that the required minimum weight shall be supported by the test specimen for 3 minutes without causing a separation of more than 1 inch. Three specimens of each interface shall be tested.

4.6.9.5.1 Test specimens. Test specimens shall be 2-1/2-inch by 8-inch strips cut longitudinally from the hose. For determining adhesion of the cover or tube to the reinforcement, a 1/4-inch wide strip of the cover or tube shall be cut and separated from the underlying reinforcement on each side of a 1-inch wide centered test strip. For determining adhesion between plies of the reinforcement, two plies shall be cut through on each side of the center test strip. The plies to be separated shall be at opposing angles.

4.6.9.6 Adhesion after immersion. Adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other, shall be determined after the test specimens (see 4.6.9.5.1) have been immersed in type III fluid of TT-S-735 for 46 hours ± 1/2 hour at 80°F ± 9°F. The test procedure for adhesion shall be method 6411 of FED-STD-601 except that the required minimum weight (10 pounds) shall be applied to the wet specimen within 1 minute after it is removed from the fluid. The required minimum weight shall be supported by the test specimen for 3 minutes without causing a separation of more than 1 inch. Three specimens of each interface shall be tested.

4.6.9.7 Nonvolatile extractable material.

4.6.9.7.1 Test specimen. Each specimen shall consist of strips about 1/16-inch square by 2 inches long, totaling about 5 grams in weight, cut from buffed pieces of the tube of the hose.

4.6.9.7.2 Procedure. The specimen shall be weighed and placed in an extraction thimble in the extraction apparatus shown in method 16001 of FED-STD-601. One hundred milliliters (ml) of type I fluid of TT-S-735 shall be added to the flask. The specimen shall be extracted for 46 hours ± 1/2 hour at such a rate that 2-1/2 to 3-1/2 minutes are required to fill and empty the extraction cup. The extract shall be poured into a weighed container. The specimen and flask shall be rinsed with 25 ml of fresh type I fluid and the wash added to the extract. The nonvolatile residue shall be determined from the extract in accordance with ASTM D381, except that the evaporation time shall be 45 minutes. Weight of the residue shall be expressed as percent of the original weight of the specimen. The test shall be performed in duplicate and the results averaged.

4.6.9.8 Phosphate plasticizer test. Phosphate plasticizer shall be determined qualitatively in accordance with method 15825 of FED-STD-601, using a small piece of clean sodium about the size of a pea, a test tube about 3 inches long, and the specimens of the residue remaining after the nonvolatile extractable material test specified in 4.6.9.7 through 4.6.9.7.2. The procedure shall be repeated if phosphorus is found, omitting the addition of the nonvolatile extractable material in order to determine whether phosphorus was present in any form in any of the reagents used.

4.6.9.9 Resistance to ozone.

4.6.9.9.1 Test specimens. Two 5-inch lengths shall be cut from the sample length of hose.

4.6.9.9.2 Apparatus. The apparatus shall consist of a test chamber with an ozone generating source outside the chamber capable of maintaining an ozone-air atmosphere in the chamber at the temperature and concentration desired; a circulating air oven, as described in method 7221 of FED-STD-601; and two pairs of metal plates, 3/8-inch by 1-1/2-inches by 8 inches each, with bolt holes spaced centrally 1/2 inch from each end, and provided with bolts and nuts for clamping the specimens.

4.6.9.9.3 Procedure. The specimen shall be placed between the metal plates and clamped so that the distance across one of the loops thus formed is 2 inches. The bulk of the specimen shall be reduced by cutting away the loop that projects from the side opposite the measured loop. Cut edges of the specimen shall be coated with paraffin to seal them from ozone. The clamped specimens shall be conditioned in an air oven for 16 to 24 hours at 100°F ± 2°F. Specimens shall then be placed in the ozone test chamber for 166 hours ± 1/2 hour at 100°F ± 2°F. Concentration of ozone in the chamber shall be 1.00 ± 0.05 parts of ozone per one million parts of air by volume. Specimens shall be examined at the end of this period for cracking of the hose cover using 7X magnification to determine conformance with the requirements specified in 3.9.

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4.6.10 Flexibility test. One end of coupled hose shall be supported from an overhead attachment. The hose shall be bent 180 degrees. A torque shall be applied to the free end of the bent configuration and the hose twisted 90 degrees in each direction.

4.7 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging, packing and marking. Hose (and hose with couplings) shall be preserved and packaged levels A or C; packed level A, B, or C as specified (see 6.2) and marked in accordance with MIL-H-775.

6. NOTES

6.1 Intended use. Hoses covered by this specification are intended for use with probe fueling system receivers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Length of coupled hose required (see 3.8).
(c) Level of preservation-packaging and level of packing required (see 5.1).

6.3 Suggested formulations. Suggested formulations for the tube, friction and cover stock of the hose are listed as follows:

Tube stock

Compounding ingredients	Parts by weight
Paracril CLT	100
Philblack N330	60
Sulfur	1.5
Protox 166	5
Stearic acid	1
Hycar 1312	10
TP-90B	10
Altax	1
Methyl tuads	0.2

Friction stock

Compounding ingredients	Parts by weight
Hycar 1041	100
Philblack N550	10
Hi Sil 233	30
Dixie clay	30
Sulfur	2
Protox 166	5
Stearic acid	1
Cumar P-10	30
Bardol B	15
Hylene MP	10
Altax	1.5

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Cover stock

Compounding ingredients	Parts by weight
Paracril OZO	100
Philblack N330	30
Regal 300	30
Sulfur	1.5
Protox 166	5
Stearic acid	1
Flexzone 6H	3
Cumar P-25	15
Plasticizer SC	10
Flexol DOP	10
Altax	1.5
Thionex	0.2

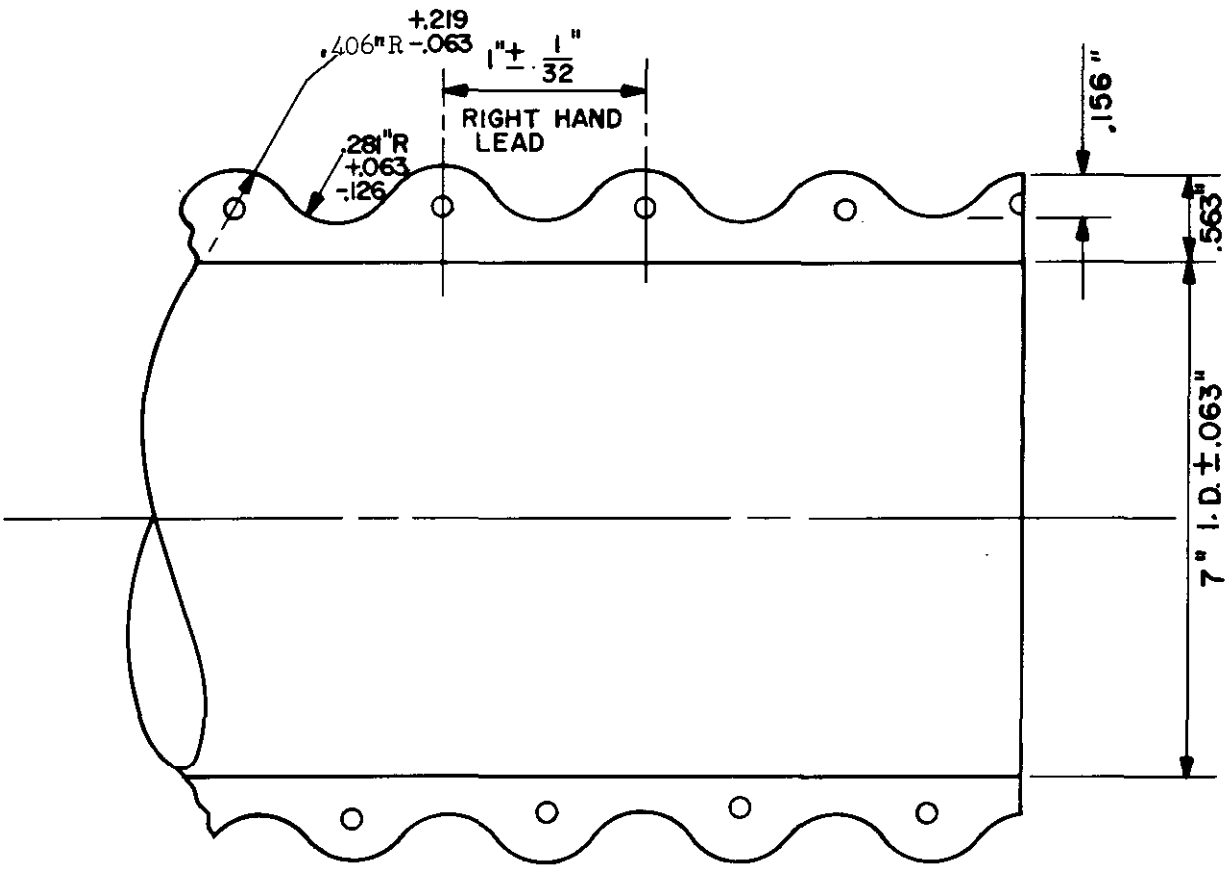
6.3.1 Certain ingredients appear as proprietary names in the formulations listed in 6.2 since these were the specific ingredients used in the development of this specification. It is not intended to limit the choice of the commercial source of an ingredient or to infer that one source is better than another.

6.3.2 The manufacturer is not obligated to use these formulations; this information is offered only as a suggestion. The use of these formulations will not relieve the manufacturer from having his product tested against the requirements of this specification.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 4730-N576)

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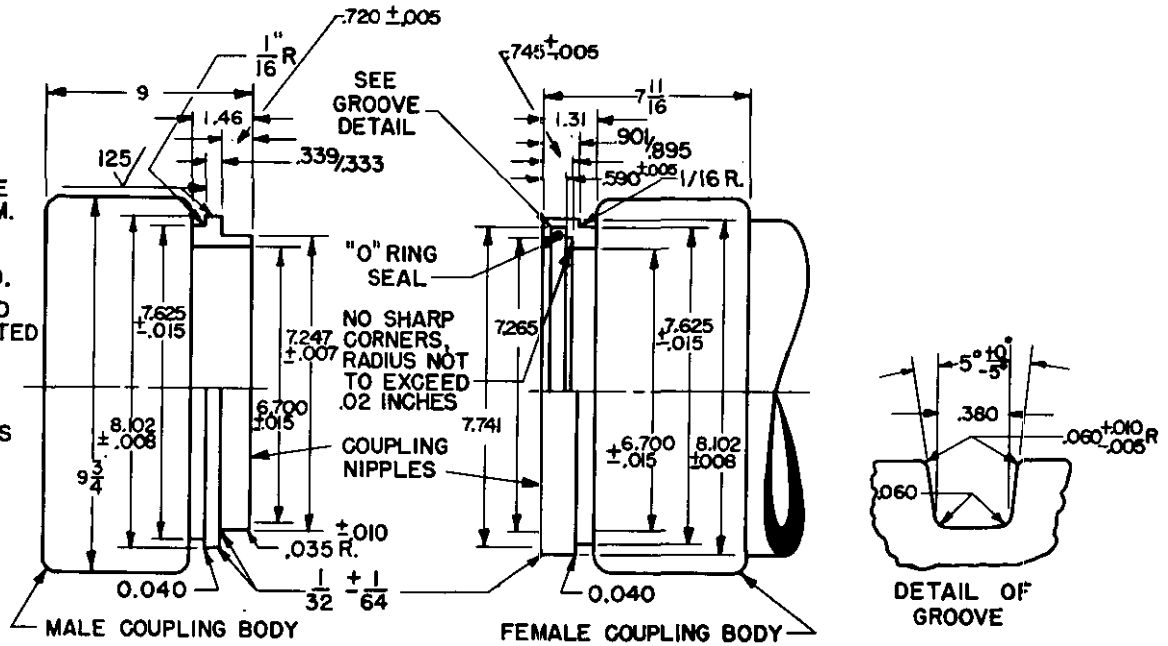
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FIGURE 1. Convolution details of hose carcass.

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NOTES:

1. ALL DIAMETERS MUST BE CONCENTRIC WITHIN .010" TIR.
2. TOLERANCES SHALL BE ±.005 ON DECIMAL DIM. ± 1/64 ON FRACTIONS WHERE NOT SPECIFIED.
3. DETAILS NOT SPECIFIED SHALL BE AS DESIGNATED BY THE SUPPLIER AND AS APPROVED BY NAVSEC.
4. ALL GROOVE SURFACES TO BE 63.

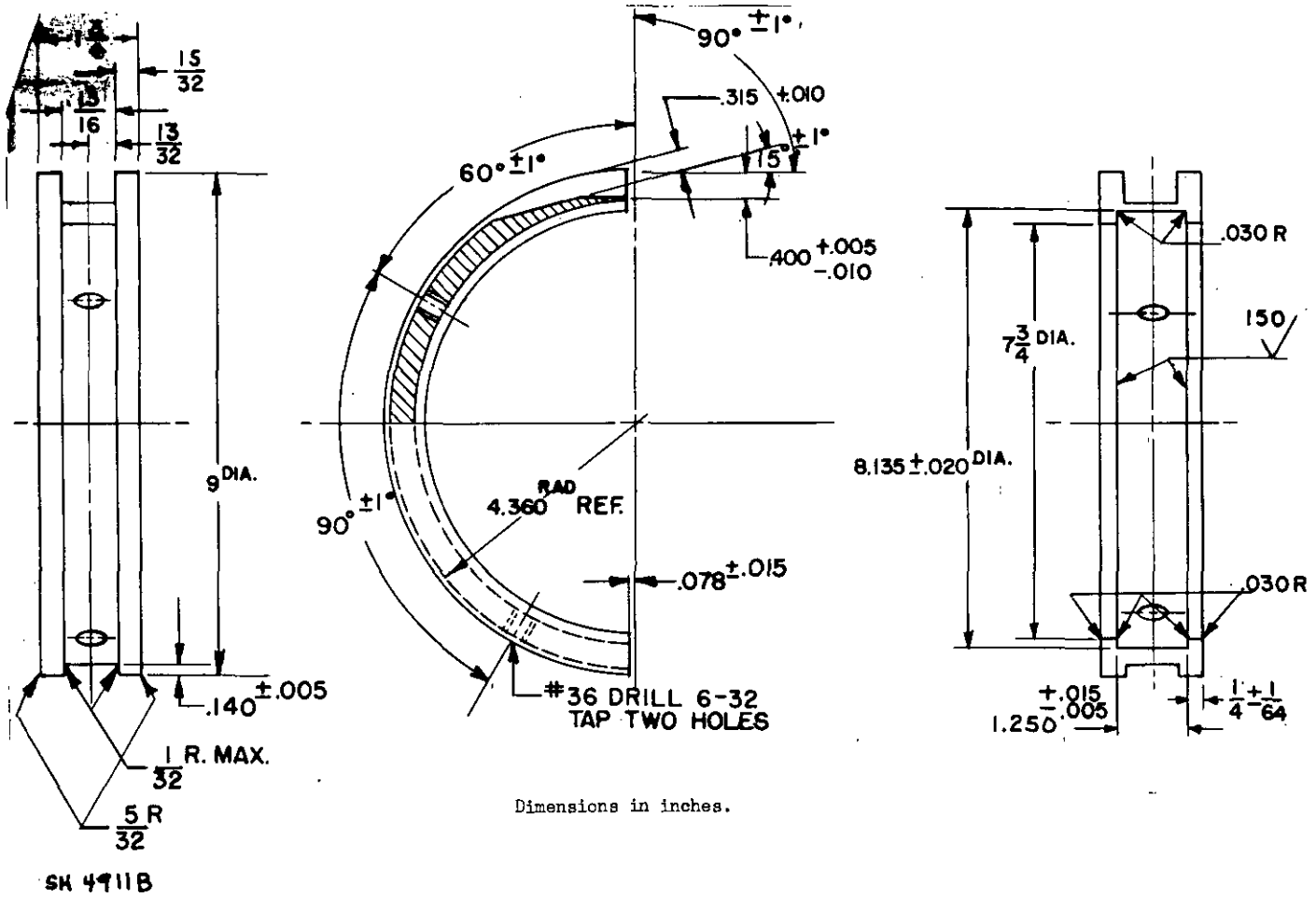


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Maximum weight of the coupling assembly - 35 pounds.
Dimensions in inches.

FIGURE 2. Reattachable coupling.

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Dimensions in inches.

FIGURE 3. Split clamp half - details.

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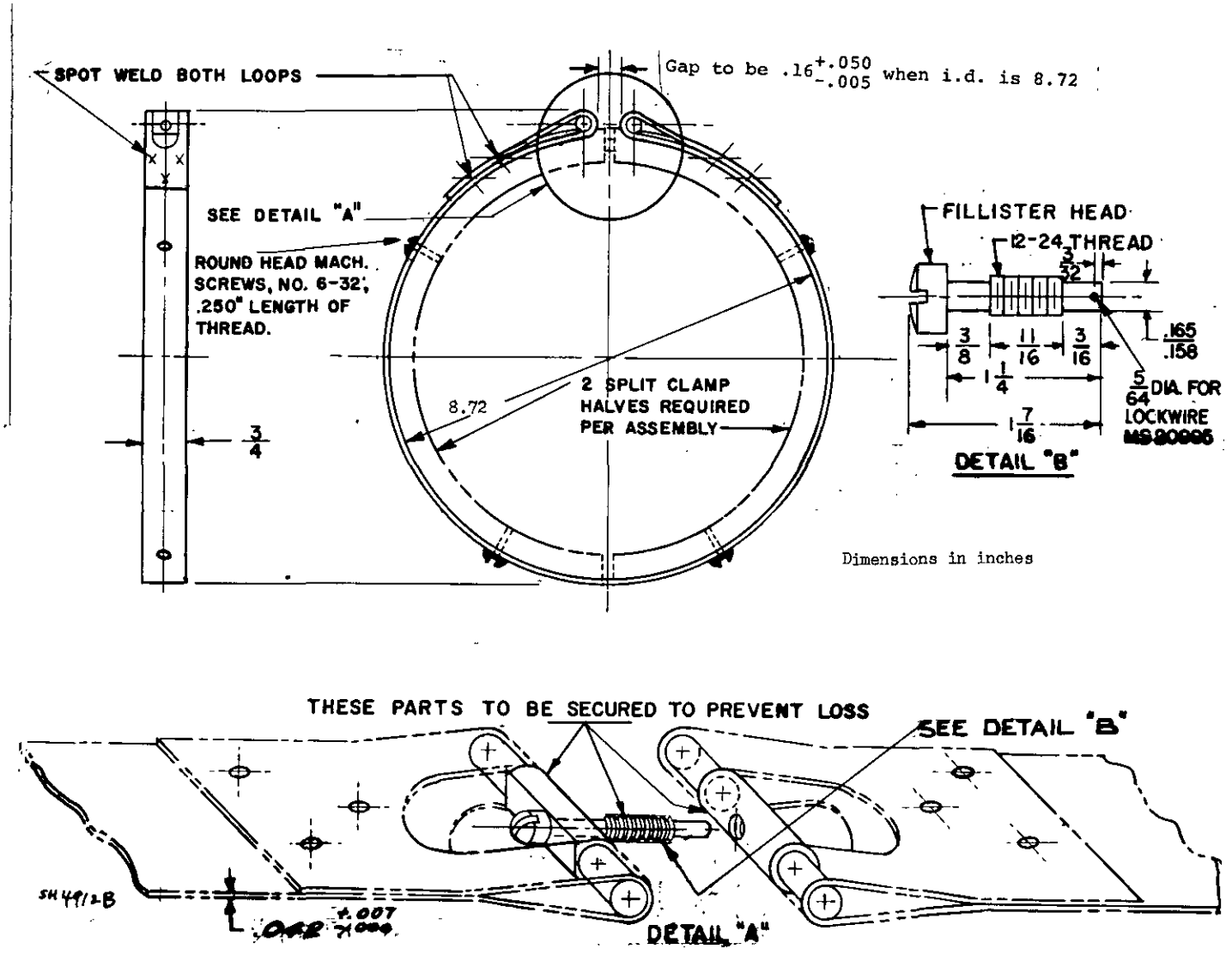


FIGURE 4. Split clamp and band - assembly and details.

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